



Patterns and Tools for Parallel Programming in C#

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Level: Intermediate

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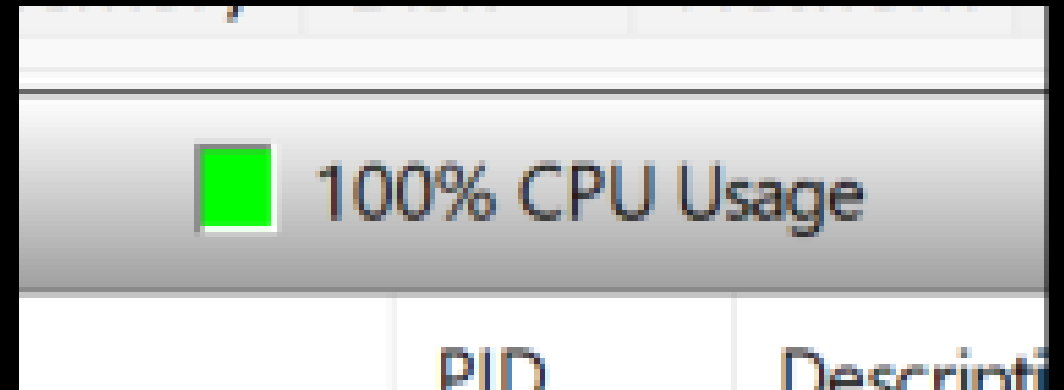
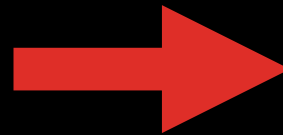
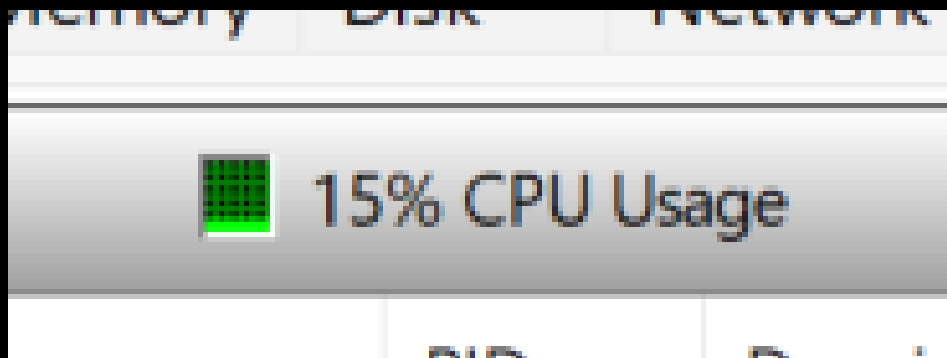
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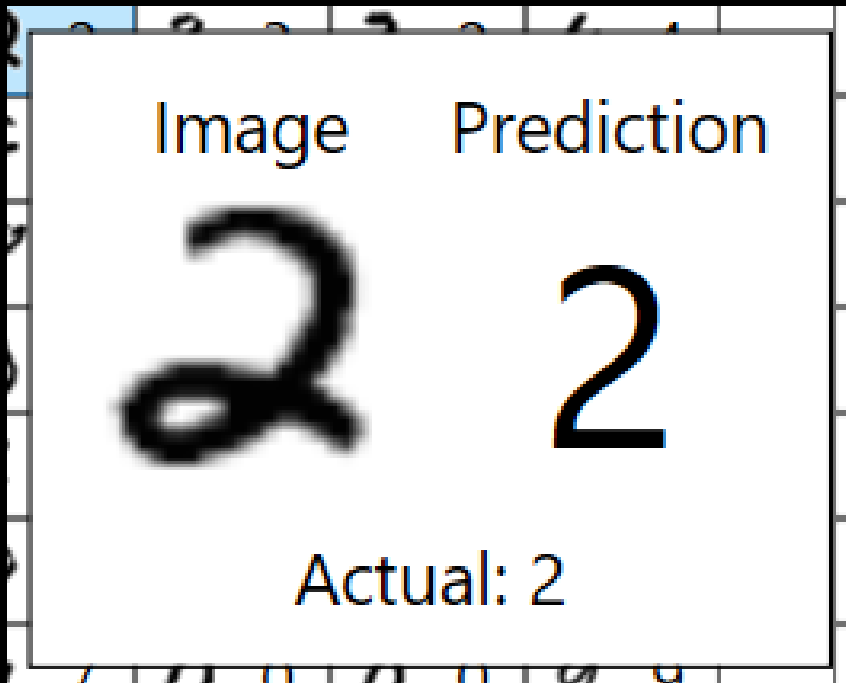
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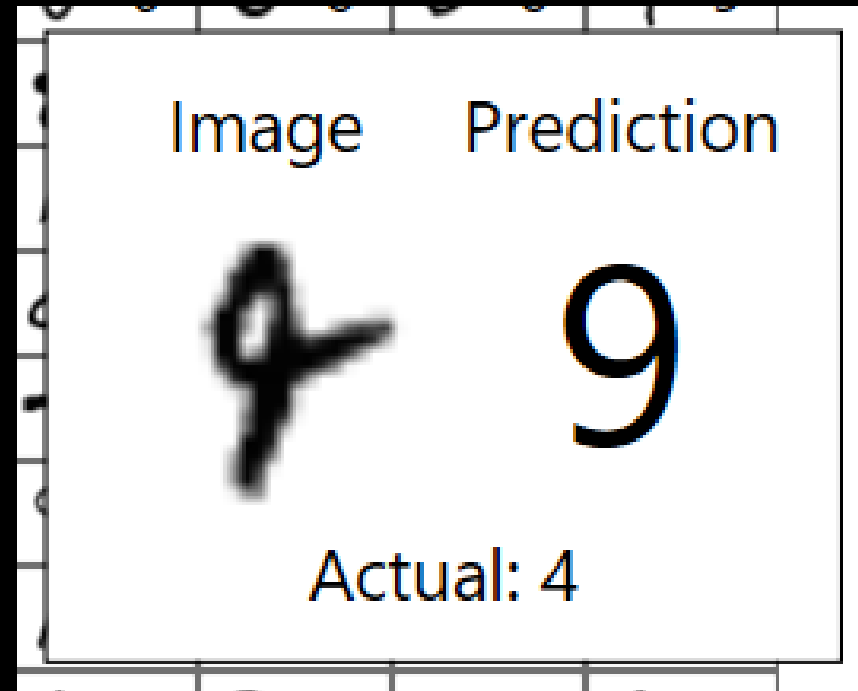
Today's Goal



Recognizing Hand-written Digits



Correct



Incorrect

Parallel Tools & Considerations

Parallel Tools

- `async / await`
- Task with continuations
- `Parallel.ForEachAsync`
- Channel

Considerations

- Parallel Continuations
- Continuing on the Main Thread
- Limiting Parallelism
- Thread-safe Updates

Comparing Parallel Approaches

	Await	Task	ForEachAsync	Channel
Runs in Parallel	No	Yes	Yes	Yes
Continuation on Main Thread	Yes	Yes (optional)	No	Yes
Continuation in Parallel	No	Yes	Yes	No (optional)
Set Degrees of Parallelism	No	No	Yes	No

await is Sequential

- Multiple "await"s run in sequence (one at a time)

```
await GetPerson(1);  
await GetPerson(2);  
await GetPerson(3);
```

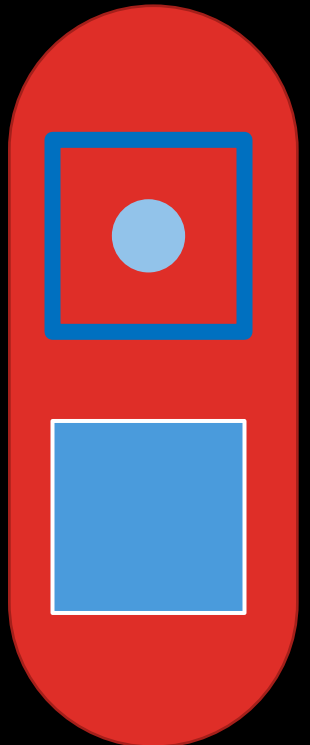
GetPerson(2) will not run until after GetPerson(1) is complete.

GetPerson(3) will not run until after GetPerson(2) is complete.

Looping await

- Each iteration at the loop with pause at the `await`
- The next iteration will not run until the previous is done

```
foreach (var imageData in validation)
{
    var prediction =
        await classifier.Predict(imageData);
    DisplayImages(prediction);
}
```



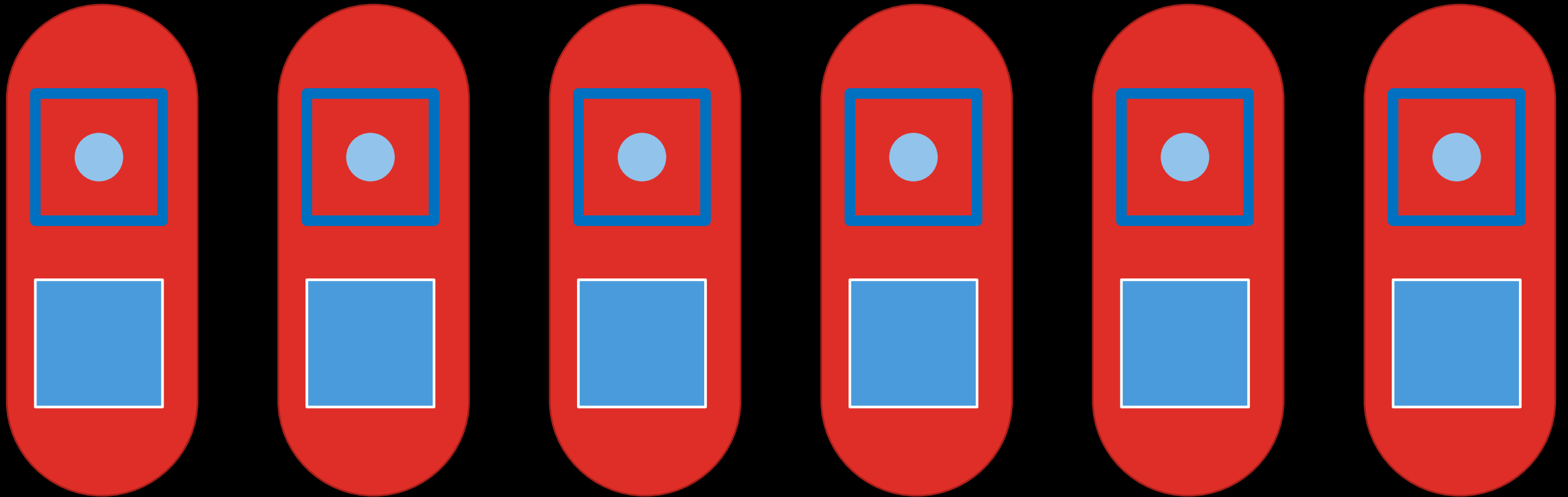
Parallel with Task

- Multiple non-awaited Tasks can run in parallel (at the same time)

```
GetPerson(1).ContinueWith(...);  
GetPerson(2).ContinueWith(...);  
GetPerson(3).ContinueWith(...);
```

GetPerson(1), GetPerson(2), and GetPerson(3) all run at the same time.

Get Data / Use Data



Parallel Loop with Task

- Task with a continuation does not pause the loop for each iteration
- The Tasks run in parallel

```
foreach (var imageData in validation)
{
    var predictionTask = classifier.Predict(imageData);
    predictionTask.ContinueWith(
        t => DisplayImages(t.Result));
}
```

Waiting for Parallel Tasks

- `await Task.WhenAll` can be used to determine when all tasks are complete.

```
List<Task> allTasks = new();  
allTasks.Add(task1);  
allTasks.Add(task2);  
allTasks.Add(task3);  
await Task.WhenAll(allTasks);
```

Parallel.ForEachAsync

- Loops over items and runs them in parallel
- “**await**” can be used safely inside the loop
 - Note: this is not true for “Parallel.ForEach”
- The entire loop can be “**await**”ed (this means all iterations will be complete)
- Available in .NET 8 / 9 / 10 (not .NET Framework)

Parallel.ForEachAsync

Waits for all iterations to finish

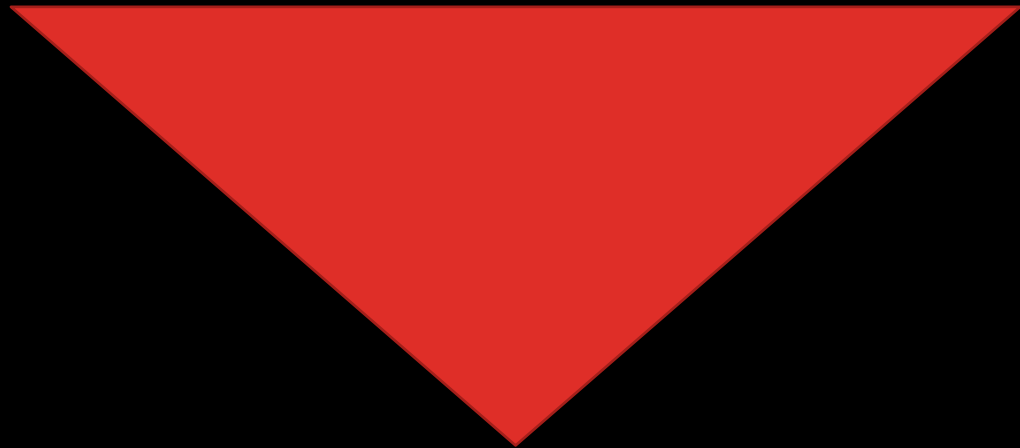
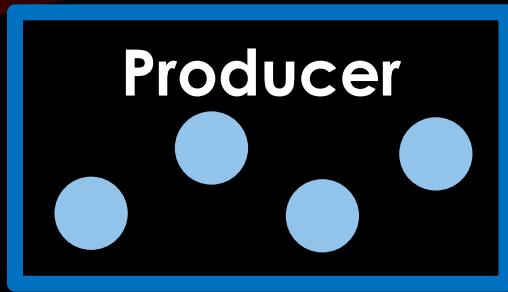
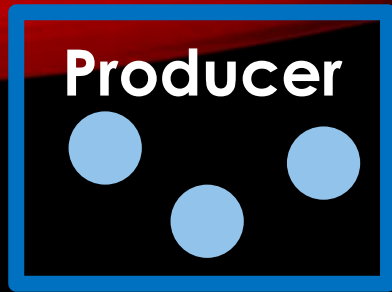
```
await Parallel.ForEachAsync(
    validation, ← The items to iterate over
    new ParallelOptions() { MaxDegreeOfParallelism = 10 },
    async (imageData, _) =>
    {
        var prediction =
            await classifier.Predict(imageData);
        DisplayImages(prediction);
    });
```

The method to run in parallel

ParallelOptions

- ParallelOptions
 - MaxDegreeOfParallelism
Limits how many iterations run at the same time.
 - CancellationToken
Cancel the loop before completion.
 - TaskScheduler
Manage where the iterations run.

Producer / Consumer



Consumer

What are Channels?

- Similar to a concurrent queue.
 - Write items to the channel.
 - Read items from the channel in the same order they were added.
 - Items are removed as they are read.
- Concurrent means that you can safely write and read from multiple threads without worry of missed writes or duplicate reads.

Where to Get Channels

- Built in to .NET 8 / 9 / 10
 - `using System.Threading.Channels;`
- .NET Framework NuGet Package
 - `System.Threading.Channels`

Parallel with Channels

- Overall Steps
 - Create a channel
 - Write to a channel
 - Read from a channel
 - Mark the channel “complete”

Creating a Channel

- `CreateBounded<T>`
 - Creates a channel of a specific size
 - If the channel is full, writers are blocked until space is available

```
var channel = Channel.CreateBounded<Person>(10);
```

Writing to a Channel

- `writer.WriteAsync()`
 - Writes an item to the channel

```
await writer.WriteAsync(person);
```


Reading from a Channel (.NET 8 / 9 / 10)

- `reader.ReadAllAsync()`
 - Returns an `IAsyncEnumerable<T>`

```
await foreach(var person in reader.ReadAllAsync())  
{  
    // use item here  
}
```

- If the channel is empty, the loop will pause until an item is available.
- If the channel is “complete”, the loop will exit.

Reading from a Channel (.NET Framework)

- `WaitToReadAsync` and `TryRead` can be used to retrieve items.

```
while (await reader.WaitToReadAsync())
{
    while (reader.TryRead(out Person person))
    {
        // use item here
    }
}
```

Marking a Channel “Complete”

- `writer.Complete()`
 - Indicates that no further items will be written
 - Writing to a “complete” channel throws an exception
 - Reading from a “complete” channel will continue normally until the channel is empty

Comparing Parallel Approaches

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Parallel Considerations

- Continuing on the main thread
- Limit degrees of parallelism
- Thread-safe collections
- Thread-safe operations

Continuations on Main Thread

- For Windows Desktop and MAUI applications, the **Dispatcher** class can be used to run a delegate on the UI thread.

Dispatcher Samples

- WPF / WinForms - Dispatcher.Invoke(Action)

```
Dispatcher.Invoke(() => CreateUIElements(prediction));
```

- WinUI3 / MAUI - Dispatcher.Dispatch(Action)

```
Dispatcher.Dispatch(() => CreateUIElements(prediction));
```

Max Degrees of Parallelism

- `SemaphoreSlim` can be used to limit the number of concurrent processes.
- Steps
 - Create `SemaphoreSlim` with initial count.
 - `await WaitAsync` to wait for entry.
 - Call `Release` when done.

Semaphore Slim Sample

```
using SemaphoreSlim semaphore = new(10);
foreach (var imageData in validation) {
    await semaphore.WaitAsync();
    var predictionTask = classifier.Predict(imageData);
    var continuation = predictionTask.ContinueWith(t => {
        CreateUIElements(t.Result);
        semaphore.Release();
    });
}
```

Thread-safe Collections

- `List<T>` is not thread-safe. If you call `Add` from concurrent processes, some items may not be added.
- When writing to a collection from concurrent processes (such as parallel code), use a concurrent collection.
- `System.Collections.Concurrent`
- `BlockingCollection<T>` is a good place to start.

Thread-safe Increment

- Increment (++) and decrement (--) operators are not thread-safe.
- The `Interlocked` class has a useful set of thread-safe methods.
 - Increment
 - Decrement
 - Add
 - Several others

Interlocked.Increment Sample

```
await Parallel.ForEachAsync(items, async (i, _) => {  
    try {  
        // Process Item  
        Interlocked.Increment(ref TotalProcessed);  
    }  
    catch (Exception ex) {  
        Interlocked.Increment(ref TotalExceptions);  
        ExceptionReporter.ShowException(ex);  
    }  
});
```

Comparing Parallel Approaches

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Other Tools

- **Parallel.ForEach**

Does not work with async in the iteration

- **PLINQ / .AsParallel()**

Does not work with async in the iteration

- **DataFlow**

<https://learn.microsoft.com/en-us/dotnet/standard/parallel-programming/dataflow-task-parallel-library>

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Thank You!

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<https://github.com/jeremybytes/vslive2025-orlando>